

Kabir V. Dabholkar

POSTDOCTORAL FELLOW, PRINCETON UNIVERSITY

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About Me

- **Research interests:** Computational/theoretical neuroscience, machine learning for science. High-dimensional models of brain-dynamics, inferred from neural/behavioral data: their analysis and evaluation. Individuality, dynamics of learning and neural representations in biological and artificial systems. Developing numerical tools based on dynamical systems, machine learning, and statistical physics.

Research Experience

With Prof Jonathan Pillow, Pillow Lab, Princeton University

Princeton NJ, USA

POSTDOCTORATE

May 2026–Present

- To be decided.

With Dr Ran Darshan, Darshan Lab, Tel Aviv University

Tel Aviv, Israel

POSTDOCTORATE

February 2026–April 2026

- Studying the geometry of neural activity and dynamics during decision making.

With Prof Omri Barak, Barak Lab, Technion

Haifa, Israel

DOCTORATE (PHD)

2021–2025

- Studying the dynamics of exploration and learning in biological and artificial systems in a space of solutions to a computational task, and the persistence of behavioral individuality in these systems despite their flexibility and susceptibility to drift.
- Refining the model-selection process for dynamical latent variable models inferred from spiking neural data, to encourage simple representations of latent structure. Analysis with Hidden Markov Models and state-of-the-art latent variable models.
- Developing tools for the analysis of RNNs using Koopman theory.
- **PhD thesis:** *From data to models to understanding*

With Prof Omri Barak, Barak Lab, Technion

Haifa, Israel

MASTER'S THESIS

2020–21

- Studied variability in recurrent neural network (RNN) models of neural computation. Developed methods to filter the space of RNN models based on behavioral perturbations.

With Prof Aurnab Ghose, Neuronal Connectivity Laboratory, IISER Pune

Pune, India

UNDERGRADUATE SUMMER PROJECT

Summer 2019

- Implemented software tools and algorithms to extract single-neuron fluorescence traces from calcium imaging data of zebrafish whole brains and slices, to study neural circuits underlying internal states of hunger and satiety.

With Prof Suhita Nadkarni, Computational Neurobiology Lab, IISER Pune

Pune, India

UNDERGRADUATE SEMESTER AND SUMMER PROJECTS

Summer 2018, Fall 2018, Spring 2019,

Spring 2021

- Worked on a spatially detailed stochastic model of the Schaffer Collateral axon in the hippocampus to study calcium signalling at the CA3 presynapse and its role in presynaptic short-term plasticity. Explored how mishandling of internal calcium stores could be implicated in neurodegenerative disease.

Publications

- Kabir Dabholkar and Omri Barak. *Separatrix Locator: Finding Separatrices with Deep squashed Koopman Eigenfunctions*. UniReps Blog (2025)
- Kabir Dabholkar and Omri Barak. *Finding separatrices of dynamical flows with Deep Koopman Eigenfunctions*. Advances in Neural Information Processing Systems 39 (2025).
- Kabir Dabholkar and Omri Barak. *When predict can also explain: few-shot prediction to select better neural latents*. arXiv:2405.14425 (2024). (accepted to PLOS Computational Biology)
- Elia Turner, Kabir Dabholkar, and Omri Barak. *Charting and navigating the space of solutions for recurrent neural networks* Advances in Neural Information Processing Systems 35 (2021).

Selected Talks and Poster Presentations

- *Finding Separatrices of Dynamical Flows with Deep Koopman Eigenfunctions* — poster, Computational and Systems Neuroscience (COSYNE), Lisbon, 2026
- *Finding Separatrices of Dynamical Flows with Deep Koopman Eigenfunctions* — contributed talk, CRSy Neuro-Cybernetics at Scale, Lisbon, 2025
- *When Predict Can Also Explain: Few-Shot Prediction to Select Better Neural Latents* — poster, Computational and Systems Neuroscience (COSYNE), Montreal, 2025
- *Characterising Differences Across Trained Recurrent Neural Networks* — poster, Computational and Systems Neuroscience (COSYNE), online, 2021
- *Generalisation in Data-Constrained Recurrent Neural Networks* — poster, Israel Society for Neuroscience (ISFN) Annual Meeting, online, 2021
- *Simulating Short-Term Plasticity with Altered Internal Calcium Stores in the Alzheimer's Presynaptic Terminal* — poster, No Garland Neuroscience (NGN), IISER Pune, 2020

Teaching Experience

Teaching Assistant, Mathematical models in Neuroscience

Faculty of Medicine, Technion

LECTURER: PROF OMRI BARAK

Winter 2021-22

- Graduate course on mathematical models in neuroscience targeted towards graduate students with an experimental background but limited mathematical and programming knowledge.
- Conducted tutorials and prepared material in the form of Jupyter notebooks.

Teaching Assistant, Neuromatch Academy: Computational Neuroscience

Online

Summer 2021

- A month-long, worldwide, virtual computational neuroscience summer school.

Teaching Assistant, Computational Approaches to Memory and Plasticity (CAMP) with a focus on dynamical neuroscience

IISER Pune

July 11-25, 2023

- Conducted tutorials on task-trained Recurrent Neural Networks (RNNs) and how to analyse them with dynamical systems. To this end, I prepared Jupyter notebooks and interactive problem solving sessions. Also prepared psychophysics experiments to compare human and RNNs on similar tasks.
- Guided student projects developed from my tutorials.

Education

5-year Integrated BS-MS at Indian Institute of Science Education and Research (IISER) Pune

2016-2021

- **1st and 2nd year** introductory courses in Mathematics, Physics, Biology, Chemistry, Earth Sciences and Social Sciences.
- **3rd and 4th year** elective courses: Neurobiology I and II, Genetics, Introductory Immunology; Numerical Analysis, Graph Theory, Algorithms; Statistical Mechanics I; Biostatistics, Data Science, Bioinformatics, Numerical Computation using MATLAB.
- **5th year Master's thesis** titled *Towards mechanistically informed Recurrent Neural Network models of neural population dynamics*, under the supervision of Prof Omri Barak (Technion).
- CGPA **9.0/10** (with distinction).

Awards and Scholarships

- **DST - INSPIRE Scholarship** 2016-2021
- **Olga and Jules Craen Young Musician of the Year 2018**